Intelligent Medical Report Analysis System

Project Domain: DL / GenAI / NLP
DSAI LAB Group 2

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Our *team*

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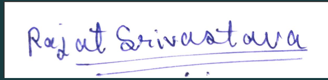

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Problem Definition (The "Why")

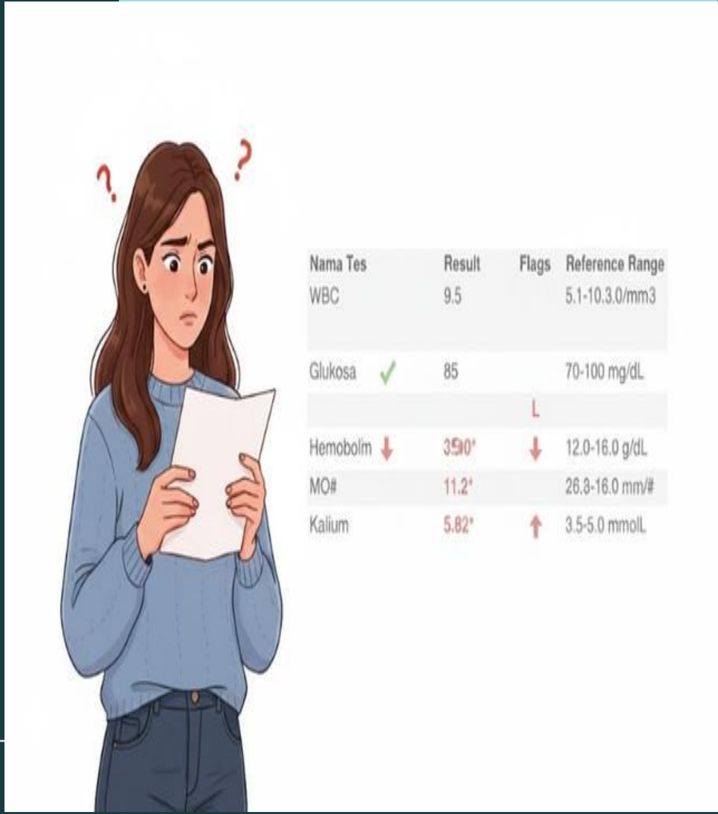
The Problem

- Medical laboratory reports contain complex clinical terminology and numerical values that are difficult for non-medical users to understand.
- Patients often struggle to identify important health indicators and determine whether their results are normal or abnormal.
- Generic AI tools can produce lengthy, unstructured explanations and may generate inaccurate or unsupported medical information (hallucinations).
- Existing hospital reports are primarily designed for healthcare professionals rather than patients.

Why It Matters

- Poor understanding of medical reports can increase anxiety and confusion.
- Patients frequently rely on repeated internet searches or follow-up questions to interpret their results.
- There is a need for a reliable, structured, and patient-friendly system that explains medical reports without replacing professional medical advice.

Develop an AI-assisted system that extracts key information from medical reports and presents accurate, easy-to-understand explanations through a structured dashboard and plain-language summary.



Project Objectives (The "What")

Steps

- 📄 **Medical Report**
- 🔍 **Extract Information**
- ⚠️ **Detect Abnormal Values**
- 💬 **Simplify Medical Terms**
- 📊 **Dashboard & Summary**

✅ In Scope	❌ Out of Scope
PDF/Image lab reports	Diagnosis
Information extraction	Prescription
Abnormality detection	Treatment advice
Dashboard & summary	Medical image analysis
Local inference	Handwritten prescriptions

Medical Info Extraction

Accurately identify laboratory values and diagnoses from uploaded reports.

→ KPI 1 : F1-score >= 0.85

→ KPI 2 : Accuracy Benchmarking

Simplified Explanations

Translate clinical jargon into plain English for non-medical users.

→ KPI 1 : Grade 8 Readability

→ KPI 2 : Flesch–Kincaid Grade Level

Local Data Privacy

Execute processing locally to ensure no data is sent to cloud servers.

→ KPI 1 : Run on 8GB RAM

→ KPI 2 : Local CPU Inference

→ KPI 3 : No cloud

Dual-Output Summary

Produce both a visual dashboard and a plain-language explanation.

→ KPI 1 : Visual Health Flags

→ KPI 2 : Structured Summary

Key stakeholders

Primary Stakeholders

- Patients
- Healthcare Students
- Medical Trainees

Secondary Stakeholders

- Doctors
- Clinicians
- Developers
- Researchers

System Stakeholders

- Hospitals & Diagnostic Labs
- Regulatory & Ethical Bodies

Patients (Non-Medical Users)

Primary end-users who upload reports and require quick, jargon-free explanations to understand vital health indicators and find abnormal limit flags easily

Medical Students & Trainees

Educational stakeholders utilizing the structured visual dashboard as a rapid case-review learning aid to quickly familiarize themselves with diverse lab layouts.

Doctors & Clinicians

Indirect secondary stakeholders who benefit from patients arriving at consultations with a reliable pre-understanding of their results, reducing repetitive explanation time.

AI Developers & Researchers

Technical stakeholders using the two-stage pipeline architecture to study local hardware limitations, lightweight inference optimization, and medical NLP pipelines.

Hospitals & Diagnostic Labs

Core institutional data providers providing anonymized real-world datasets (like clinical notes and transcriptions) required for training and validating model boundaries.

Literature Review Findings

Key Findings

Current medical AI and clinical knowledge systems show strong capabilities in medical reasoning and information storage, but they have important limitations for patient-facing report understanding.

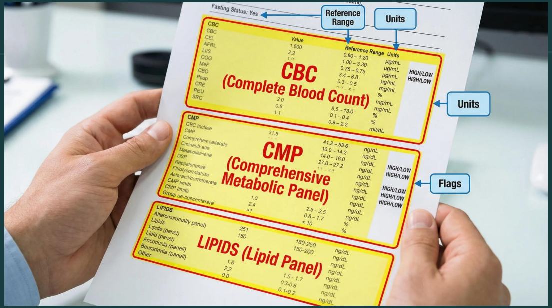
What Existing Systems Do Well

- Provide **high-quality and accurate medical knowledge**
- Support **clinical decision-making for professionals**
- Offer **trusted, evidence-based information sources**

Key Limitations Identified

- Do not automatically interpret uploaded medical reports
- Require medical training or clinical background to use effectively
- Generate long, complex explanations instead of structured summaries
- Lack patient-friendly visualization and simplification
- Not designed specifically for non-expert users

Existing Solutions



	Generic AI Platforms	Med-PaLM	Glass Health	Clinical References
FEATURES	• Broad, free-text general conversational interface. • Wide global availability.	• Domain-specific large language model. • Medically trained question answering.	• Clinical workflow automation. • Diagnostic reasoning generation.	• List the features of this product or service • Manual keyword searchable engines.
STRENGTHS	• Easy to navigate for ordinary individuals. • Wide global availability.	• State-of-the-art medical reasoning accuracy. • Medically trained question answering.	• Deep differential clinical decision support. • Diagnostic reasoning generation.	• Peer-reviewed, deeply trusted evidence databases. • Manual keyword searchable engines.
WEAKNESSES	• High risk of factually hallucinating lab values • Lacks clean formatting or scorecards.	• Completely unavailable for local, private deployment. • Lacks direct report parsing.	• Designed entirely for doctors, not patient comprehension.	• Requires active medical training to use effectively. • No automated report translation.

THE SOLUTION

An AI-Assisted System Bridging Patient Comprehension and Clinical Accuracy

01
Core Benefit

Plain-Language Clarification: Translates complex medical terminology into direct, plain English at a Grade 8 readability level or below. This instantly relieves patient anxiety and clarifies vital health indicators without forcing users to sift through unstructured text outputs.

02
Key Differentiator

Deterministic Splitting Strategy: Unlike standard generative AI models that carry a severe risk of hallucinating or altering medical figures, our system completely isolates laboratory text and numerical extraction into a separate pipeline governed by rule-based validation rules. *The AI cannot invent or alter metrics.*

03
Target Audience

Patients (Non-medical users): Who get an intuitive visual scorecard dashboard flagging abnormal markers alongside a straightforward summary paragraph.

04
Scope

Local-First Privacy Architecture: Uses lightweight, context-aware transformer models (ClinicalBERT/BioBERT) optimized to process reports locally in-memory on a standard 8GB RAM laptop. This setup ensures maximum patient data privacy because sensitive health files never touch third-party cloud servers.

VALUE PROPOSITION

Our system empowers patients by translating complex, anxiety-inducing clinical jargon into clear, actionable health summaries. We distinguish ourselves from generic AI by using **Deterministic Splitting** to ensure 100% numerical accuracy, delivering a secure, local-first experience that protects patient privacy while maintaining professional-grade clinical standards.

BUSINESS MODEL

Our open-source framework is designed as a proof-of-concept implementation to deliver high clinical utility with minimal infrastructure costs. We leverage optimized biomedical language models to enable patient accessibility and trainee education, while ensuring total data confidentiality through an entirely local, client-side inference pipeline.

Cost structure

Potential Major Costs:

- → Cloud compute resources
- → Dataset acquisition, storage, and cleaning
- → Voluntary academic research hours

Revenue streams

Outline of revenue generation plan for project sustainability:

- → Open-source research grants & academic funding
- → Custom-branded patient summary portals for diagnostic labs

Key activities

Fine-tuning base domain models (ClinicalBERT/BioBERT) , engineering deterministic validation rule scripts , optimizing local runtime footprints, and conducting readability evaluation benchmarking.

Key partnership

Hospitals, public diagnostic laboratories, and health checkup centers acting as core clinical data providers for model testing and validation datasets.

Customer segments

- → Ordinary patients and health-conscious individuals seeking jargon-free explanations.
- → Medical trainees and healthcare students
- → Developers and biomedical researchers

OPERATIONAL PLAN

Our implementation roadmap for training, local profiling, and evaluation strategy

Data Assembly

Aggregate and clean open-source clinical notes, abstract sequences, and diagnostic transcripts.

01

Model Fine-Tuning

Offload heavy transformer computation to cloud resources to bypass local hardware limits.

02

Local Hardware Optimization

Optimize the final inference models to run locally on a client-side setup

03

Evaluation & Review

Benchmark the pipeline against established biomedical NLP translation and extraction metrics.

04

Thank you